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ON

Intelligent Automation: Integration of AI-Powered Chatbots and RPA in Enterprise Applications

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Intelligent Automation: Integration of AI-Powered Chatbots and RPA in Enterprise Applications"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

This project is based on the concept of intelligent automation, where AI-powered chatbots are integrated with Robotic Process Automation (RPA) to improve system efficiency. In today's digital environment, organizations are continuously looking for ways to reduce manual work and speed up their operations. Chatbots are useful for interacting with users and understanding their queries, while RPA is effective in handling repetitive backend tasks such as data entry and transaction processing. However, when used separately, both technologies have certain limitations.

The proposed system combines these two technologies into a single framework. The chatbot is responsible for communicating with the user and identifying their intent, while the RPA component executes the required task in the backend. This integration helps in creating a smooth and automated workflow from user request to task completion.

The system was tested on different scenarios, and the results showed improvements in accuracy, response time, and overall task completion. It also reduced the need for human intervention in routine processes. This makes the proposed solution suitable for real-world enterprise applications where efficiency and scalability are important.

INTRODUCTION

With the rapid growth of technology, businesses are increasingly shifting towards automation to improve their performance and reduce manual effort. In many organizations, a lot of time is still spent on repetitive tasks like data entry, responding to customer queries, and processing requests. This not only slows down the overall workflow but also increases the chances of human error. To overcome these challenges, intelligent automation is being widely adopted.

Intelligent automation is an approach that combines multiple technologies such as Artificial Intelligence (AI) and Robotic Process Automation (RPA) to create smarter systems. These systems are capable of handling tasks with minimal human involvement while maintaining accuracy and speed.

AI-powered chatbots are commonly used for interacting with users. They can understand human language using natural language processing and provide responses accordingly. They are helpful in handling customer queries, guiding users, and providing quick support. On the other hand, RPA is used to automate repetitive backend tasks like filling forms, updating databases, and processing transactions. It works based on predefined rules and is highly efficient for structured tasks.

However, both technologies have their own limitations. Chatbots are good at understanding user queries but cannot perform actual backend operations. Similarly, RPA can execute tasks efficiently but cannot understand unstructured human language. Because of this, many processes still require human intervention, which reduces the overall efficiency of the system.

To solve this problem, there is a need to integrate both technologies into a single system. This project focuses on combining AI chatbots with RPA so that the system can first understand the user's request and then automatically perform the required task. This integration helps in creating a complete end-to-end automation solution.

METHODOLOGY

This research follows an experimental methodology to evaluate the integration of AI-powered chatbots and Robotic Process Automation (RPA) in enterprise applications.

The study uses a dataset of 2,400 annotated customer interaction transcripts collected from MultiWOZ 2.4, BANKING77, and synthetic call center data. Each transcript was manually labeled into twelve intent categories, and the dataset was split into 80% training and 20% testing for model development and validation.

For intent classification, the BERT-base-uncased model was trained using appropriate parameters such as learning rate, batch size, and dropout to improve accuracy and handle class imbalance. The integrated system was then compared with two baseline approaches: a rule-based system and a standalone chatbot without RPA integration.

The performance of the proposed framework was evaluated using key metrics including intent accuracy, task completion rate, response latency, manual intervention rate, and workflow error rate. This methodology helps assess the efficiency, reliability, and scalability of the intelligent automation framework in enterprise environments.

TOOLS AND TECHNOLOGY

The proposed intelligent automation framework uses a combination of tools and technologies to support conversational intelligence, process automation, and seamless enterprise integration. The system uses the BERT-base-uncased model for intent classification and understanding user requests, along with Natural Language Processing (NLP) techniques such as tokenization, Named Entity Recognition (NER), text preprocessing, and intent detection for extracting meaningful information from user inputs. For voice-based interactions, a Conformer-based Automatic Speech Recognition (ASR) model is used to convert spoken input into text.

For task execution, the framework uses Robotic Process Automation (RPA) bots to automate repetitive backend processes such as account management, billing operations, complaint handling, and service activation. The system supports both API-based integration and UI automation for interacting with enterprise applications, especially legacy systems. Communication between different layers of the architecture is managed using JSON-based workflow dispatching, which enables smooth data exchange and process triggering. Model training and computational processing were supported using an NVIDIA A100 GPU, while benchmark datasets such as MultiWOZ 2.4 and BANKING77 were used for training and evaluation of the system.

IMPLEMENTATION

The implementation of the proposed intelligent automation framework is based on integrating AI-powered chatbots with Robotic Process Automation (RPA) to create a unified system capable of understanding user requests and executing backend processes automatically. The implementation follows a three-layer architecture, where each layer performs a specific function and communicates with the others through well-defined interfaces. This modular structure improves flexibility, scalability, and ease of integration with enterprise applications.

The first layer is the Conversational Interface Layer, which acts as the entry point for users to interact with the system. Users can submit requests through text or voice using channels such as web chat, mobile applications, internal portals, or call center interfaces. For voice-based interactions, a Conformer-based Automatic Speech Recognition (ASR) model converts spoken input into text. Once the input is received, the system processes the user message and maintains context through a dialogue state manager, which helps understand follow-up queries based on previous interactions. This layer ensures that user requests are captured accurately and passed to the next stage for processing.

The second layer is the Intelligence and Orchestration Layer, which is responsible for understanding and routing user requests. At this stage, the input first goes through text preprocessing techniques such as tokenization, punctuation normalization, stop-word removal, and case folding to prepare the data for analysis. After preprocessing, Named Entity Recognition (NER) is applied to identify important entities such as account numbers, dates, product names, and transaction details from the user request. These extracted entities serve as parameters for automated workflows.

Once the data is processed, the system performs intent classification using the BERT-base-uncased model, which identifies the purpose of the user request from predefined intent categories such as account inquiry, billing dispute, service activation, complaint registration, or refund processing. The model also generates a confidence score for the prediction. If the confidence level is above the defined threshold, the request is accepted and mapped to the corresponding

workflow. If the confidence score falls below the threshold, the request is transferred to a human agent for review to maintain system reliability and reduce classification errors. This fallback mechanism also supports continuous improvement, as low-confidence cases can later be used for retraining the model.

After intent verification, the request is forwarded to the Workflow Dispatcher, which converts the classified intent and extracted entities into a structured JSON payload and routes it to the appropriate backend process. A configurable mapping table is used to connect each intent with its corresponding RPA workflow. This mapping approach makes the system flexible, as new intents or workflows can be added without changing the entire architecture. The orchestration layer therefore acts as the central control mechanism of the intelligent automation system.
The third layer is the Automated Execution Layer, where actual process automation takes place using RPA bots. These bots are organized into functional pools such as Account Management, Billing, Service Provisioning, Complaint Handling, and Knowledge Base Retrieval. Based on the dispatched workflow, the bots perform the required backend operations by interacting with enterprise systems. Where APIs are available, bots use API-based integration for efficient execution, while UI automation is used as a fallback for legacy applications lacking programmatic access. This allows the framework to operate across modern as well as older enterprise systems.

Once a task is executed, the result is returned in structured form and sent back to the orchestration layer. The system then converts the execution result into a natural language response, which is delivered to the user through the chatbot interface. If any workflow failure occurs, the monitoring mechanism detects the issue and triggers corrective action or escalation rather than leaving the request unresolved. This ensures reliable end-to-end process execution.
The implementation also includes model training and evaluation components to improve system performance. The intent classification model was trained using a dataset of 2,400 annotated customer interaction transcripts, collected from MultiWOZ 2.4, BANKING77, and synthetic call center data. Training was performed using defined parameters such as learning rate, batch size, and dropout, while weighted loss was used to handle class imbalance. Computational processing was supported using an NVIDIA A100 GPU, enabling efficient model training and performance optimization.

To validate the implementation, the system was tested against two baseline approaches: a traditional rule-based system and a standalone chatbot without RPA integration. The implemented framework was evaluated using metrics such as intent accuracy, response latency, task completion rate, workflow success rate, and manual intervention rate. The results showed that integrating conversational AI with RPA significantly improved both efficiency and automation performance compared to standalone systems.

Overall, the implementation demonstrates how conversational intelligence and process automation can be combined into a unified enterprise solution. By integrating user interaction, intelligent decision-making, and automated execution into a single framework, the proposed system enables scalable, reliable, and efficient intelligent automation for enterprise applications.

MAJOR FINDINGS/OUTCOMES/OUTPUT/RESULTS

The experimental evaluation of the proposed intelligent automation framework showed significant improvements over traditional automation approaches. The integrated system achieved an intent classification accuracy of 94.7%, which was higher than both the rule-based system and the standalone chatbot model. Similarly, the task completion rate increased to 89.1%, demonstrating the effectiveness of combining conversational AI with RPA for end-to-end automation.

One of the major outcomes observed was a substantial improvement in system efficiency. The average response latency was reduced from 18.4 seconds to 6.2 seconds, showing faster response generation and workflow execution. In addition, the manual intervention rate decreased from 38.6% to 9.4%, indicating that a large portion of routine customer interactions could be handled automatically without human support. The workflow error rate was also reduced to 3.1%, reflecting reliable execution performance of the automated system.

The results further showed strong performance across different intent categories. Requests such as Account Inquiry and Escalation Request achieved high accuracy, while categories like Technical Support and General Inquiry highlighted areas for future improvement. Component-level evaluation also showed effective performance of the ASR module, intent classifier, workflow dispatcher, and RPA execution bots, with a 97.2% workflow success rate.

Another important finding was the system's scalability under concurrent load. The framework maintained stable performance up to 250 concurrent sessions, with only moderate latency increase under higher loads. The study also indicated potential cost-efficiency benefits, with an estimated reduction in operational costs due to decreased manual handling and higher automation coverage. Overall, the findings confirm that integrating AI-powered chatbots with RPA improves accuracy, speed, scalability, and efficiency in enterprise automation environments

CONCLUSION AND FUTURE SCOPE

The proposed intelligent automation framework demonstrates that integrating AI-powered chatbots with Robotic Process Automation (RPA) can significantly improve enterprise process automation. By combining conversational intelligence with automated workflow execution, the system is able to understand user requests, trigger appropriate backend processes, and deliver responses efficiently. The experimental results showed improvements in intent accuracy, task completion, response latency, and reduction in manual intervention, proving the effectiveness of the proposed approach over traditional rule-based and standalone chatbot systems.

The study concludes that intelligent automation can enhance operational efficiency, scalability, and service quality in enterprise applications, especially in domains such as call center automation where repetitive processes and conversational interactions are common. The three-layer architecture also provides flexibility for integrating new workflows, improving maintainability, and supporting enterprise-level deployment. Overall, the research confirms that combining AI and RPA offers greater benefits than using either technology independently.

For future scope, the framework can be extended by incorporating continuous learning and model retraining to improve performance over time. Advanced features such as multimodal inputs (for example, text with screenshots or documents), improved explainability for routing decisions, and support for additional domains like healthcare, finance, and scheduling can further enhance the system. Future work can also focus on improving scalability through containerized bot deployment and exploring federated learning approaches for secure model improvement without centralizing sensitive data.

REFERENCES

- Adamopoulou, E., & Moussiades, L. “An Overview of Chatbot Technology,” 2020.
- Willcocks, L. P., Lacity, M. C., & Craig, A. “Robotic Process Automation: Strategic Transformation Lever for Global Business Services,” 2017.
- Syed, R. et al. “Robotic Process Automation: Contemporary Themes and Challenges,” 2020.
- Chakraborti, T. et al. “From Robotic Process Automation to Intelligent Process Automation: Emerging Trends,” 2020.
- Xu, A. et al. “A New Chatbot for Customer Service on Social Media,” 2017.
- Liu, B., & Lane, I. “Attention-based Recurrent Neural Network Models for Joint Intent Detection and Slot Filling,” 2016.

Appendices

The appendix includes supplementary information supporting the implementation and evaluation of the proposed intelligent automation framework. This includes details related to the annotated dataset of 2,400 customer interaction transcripts, intent category definitions, and sample user requests used for model training and testing. Additional supporting material such as comparative performance tables, scalability results, and component-level evaluation metrics are also included as reference material.

The appendix may also contain supporting diagrams of the three-layer system architecture, workflow mapping examples, sample JSON dispatch structures, and examples of RPA bot execution processes. These supplementary materials provide additional clarity regarding system design, experimental setup, and implementation details that support the findings presented in the report.